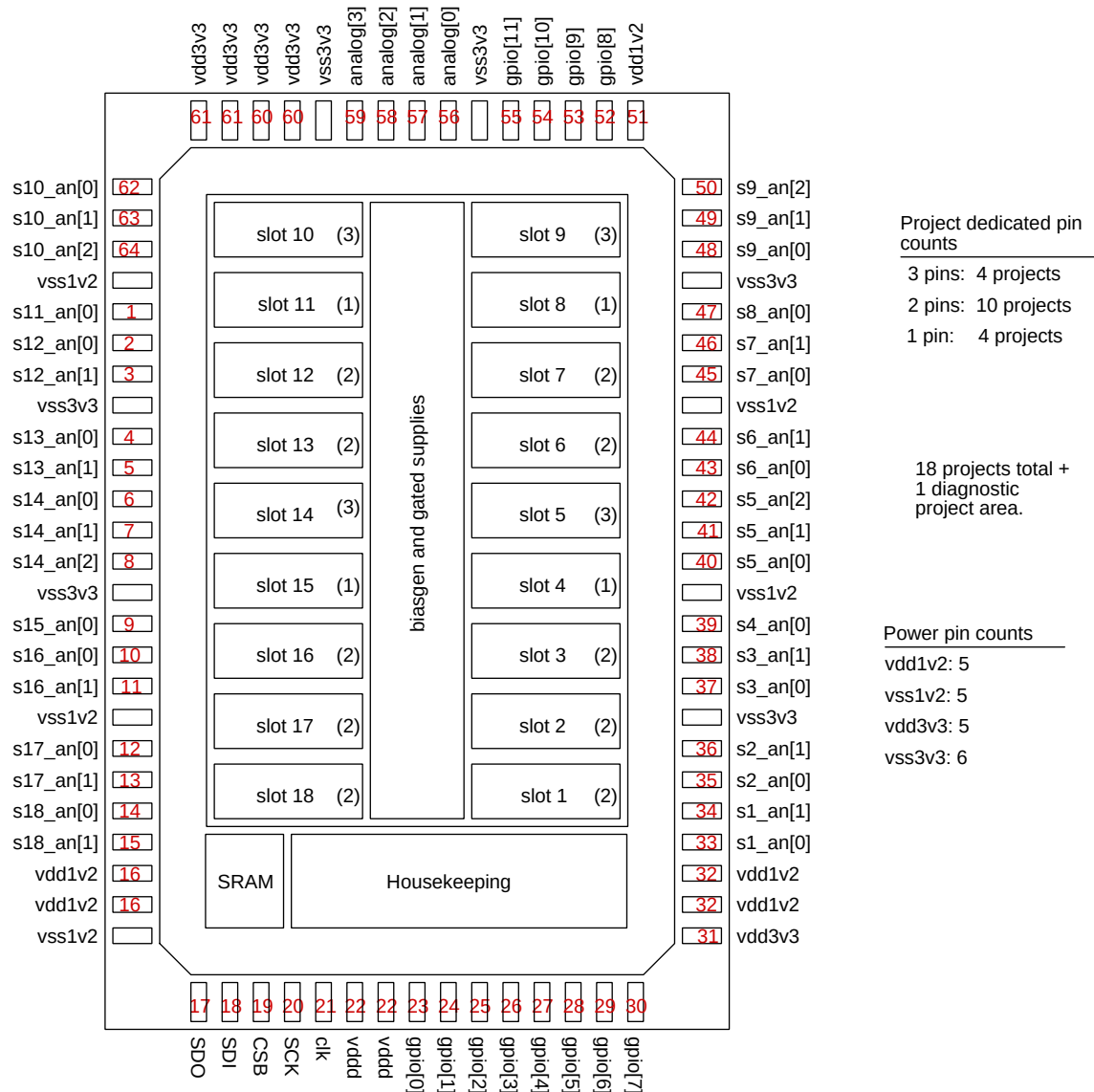


Analog harness redesign for Chipalooza #2 (80 pin padframe in 64 pin QFN package)



Pins marked in red are QFN package pins (see bonding diagram). All ground pins are downbonded to the QFN paddle. Each power supply has one double-bonded pin. Each project with one or more dedicated pins has a pin adjacent to a ground return.

Pinout for Chipalooza #2 analog harness (64 pin QFN)

s1_an[1:0] Project slot 1 dedicated analog I/O
s2_an[1:0] Project slot 2 dedicated analog I/O
s3_an[1:0] Project slot 3 dedicated analog I/O
s4_an[0] Project slot 4 dedicated analog I/O
s5_an[2:0] Project slot 5 dedicated analog I/O
s6_an[1:0] Project slot 6 dedicated analog I/O
s7_an[1:0] Project slot 7 dedicated analog I/O
s8_an[0] Project slot 8 dedicated analog I/O
s9_an[2:0] Project slot 9 dedicated analog I/O
s10_an[2:0] Project slot 10 dedicated analog I/O
s11_an[0] Project slot 11 dedicated analog I/O
s12_an[1:0] Project slot 12 dedicated analog I/O
s13_an[1:0] Project slot 13 dedicated analog I/O
s14_an[2:0] Project slot 14 dedicated analog I/O
s15_an[0] Project slot 15 dedicated analog I/O
s16_an[1:0] Project slot 16 dedicated analog I/O
s17_an[1:0] Project slot 17 dedicated analog I/O
s18_an[1:0] Project slot 18 dedicated analog I/O

All project slot dedicated analog I/O can be swapped out for digital I/O per request.

analog[3:0] Shared analog I/O

gpio[11:0] Shared digital I/O

clk Sequencer external clock input

SCK SPI clock

CSB SPI enable

SDI SPI data in

SDO SPI data out

vdd3v3 Project 3.3V and global ESD domain power supply

vdd1v2 Project 1.2V domain power supply

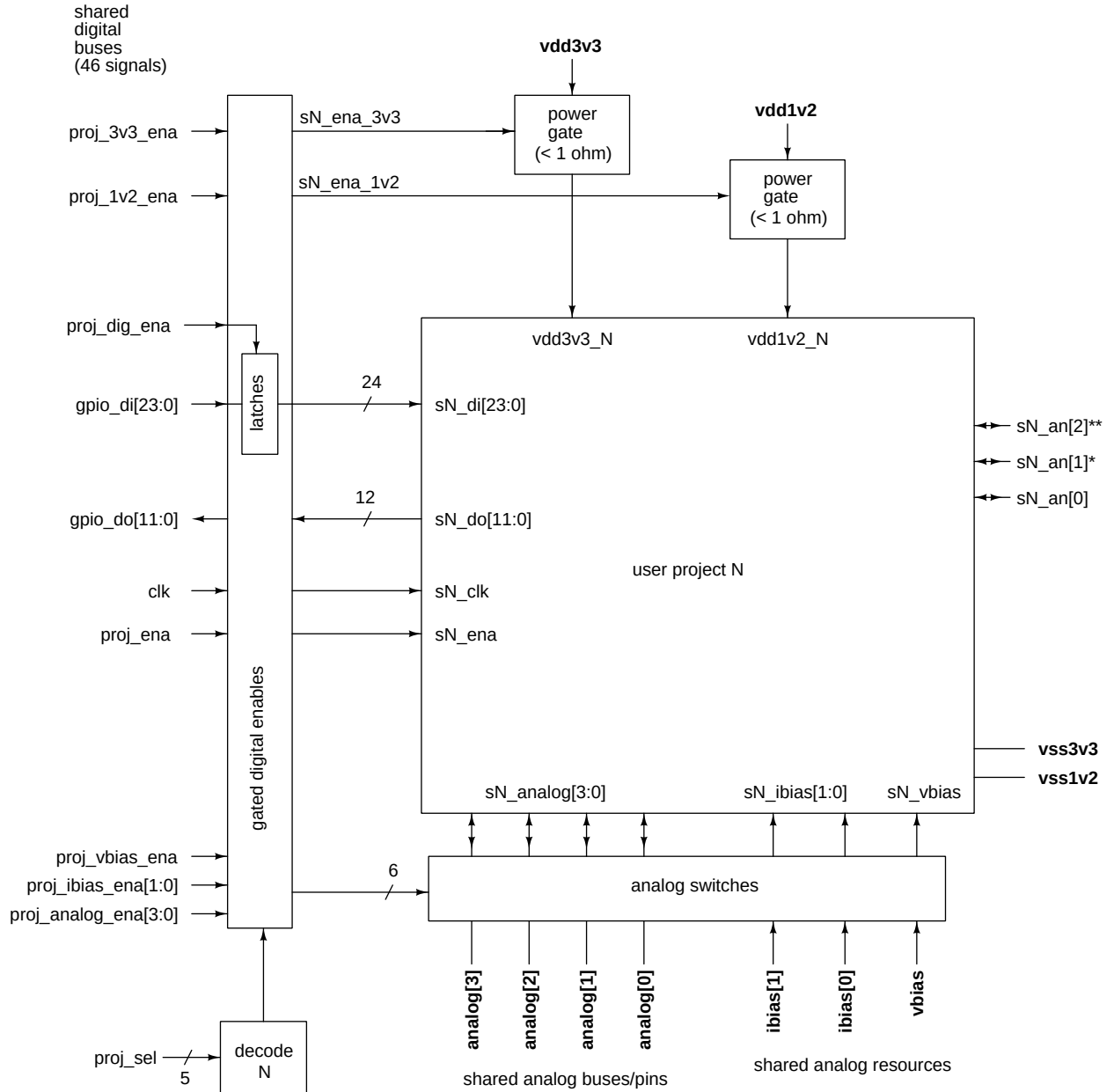
vddd Global/housekeeping 1.2V domain power supply

vss3v3 Project 3.3V and global ESD domain ground return

vss1v2 Project and housekeeping 1.2V domain ground return

All ground returns are downbonded to the QFN paddle

Block diagram for Chipalooza #2 analog harness user project wrapper



NOTES:

All input signals are held at zero when the project is not selected. All output signals are held at high impedance when the project is not selected.

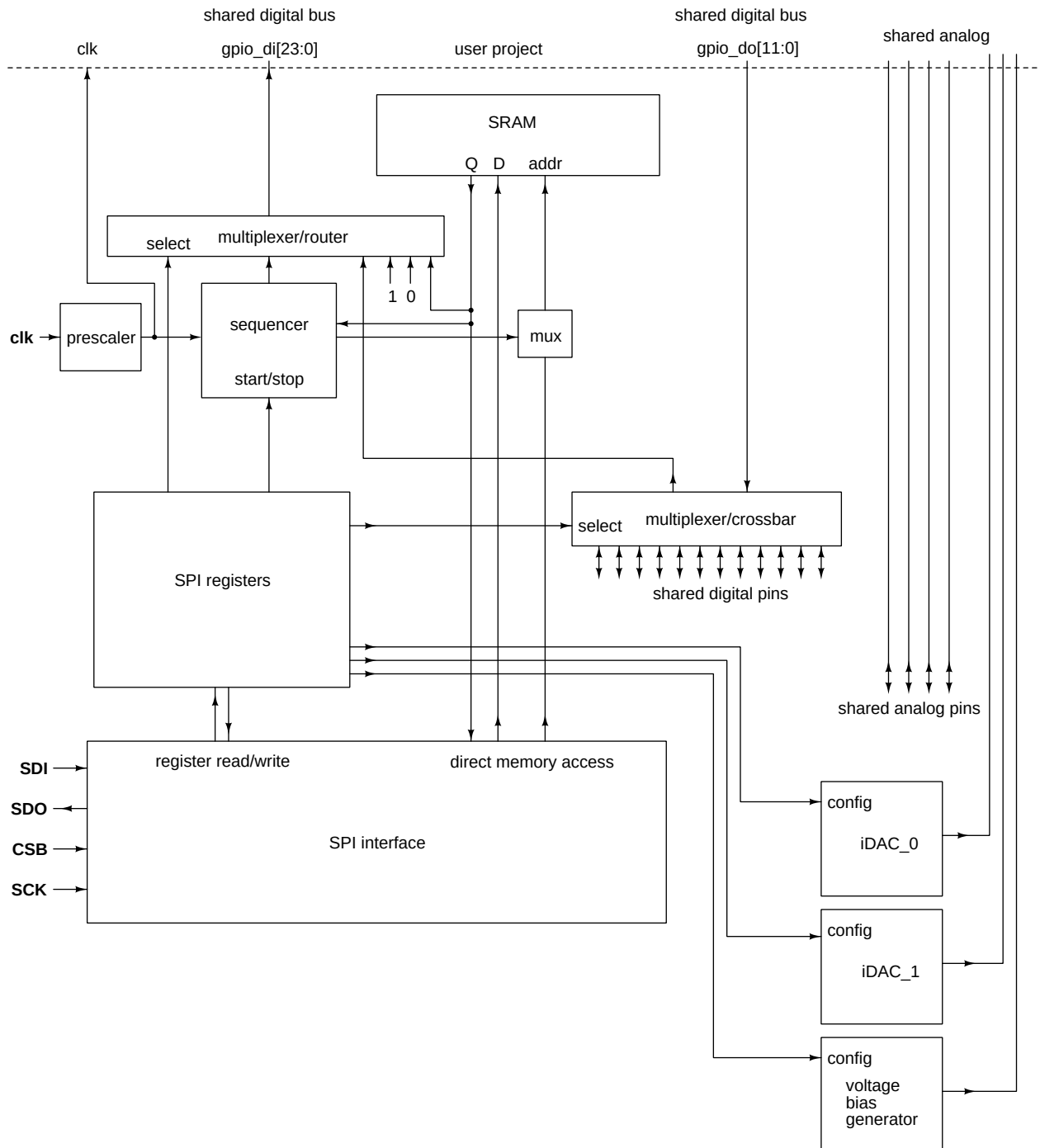
*sN_an[1] is available on fourteen of the project slots and must be approved to be used.

**sN_an[2] is available on four of the project slots and must be approved to be used.

The vdd3v3 pin for the user project is independent of the vddio pin for ESD and I/O. It can be set to any value the user project requires.

The vdd1v2 pin can also run the digital at 1.5V. Note that this supply also drives the SPI and sequencer.

Block diagram for Chipalooza #2 analog harness housekeeping and sequencer



Chipalooza #2 analog harness housekeeping and sequencer registers

SPI commands

0x00 No operation
0x80 Write until CSB raised
0x40 Read until CSB raised
0xC0 Simultaneous read/write until CSB raised
0x20 Write SRAM until CSB raised
0x10 Read SRAM until CSB raised
0x30 Simultaneous read/write SRAM until CSB raised
0x01 Run sequencer, looping
0x02 Run sequencer, single-shot
0x03 Stop sequencer
0x04 Reset digital

SPI registers (8 bits each):

0x00 Reserved/unused
0x01 Manufacturer ID, low byte (fixed field)
0x02 Manufacturer ID, high byte (fixed field)
0x03 Product ID (fixed field)
0x04 to 0x07 Mask ID (via programmed on 32-bit ROM)
0x0E Sequencer clock prescaler
0x0F Pattern generator clock prescaler
0x10 Sequencer mode (low 3 bits)
0x11 SRAM mode (low 2 bits)
0x12 to 0x13 Sequencer start count (16 bits)
0x14 to 0x15 Sequencer stop count (16 bits)
0x16 to 0x17 Pattern generator stop value (10 bits)
0x18 bias generator coarse/fine control (low 4 bits)
0x19 iDAC 1 value (low 5 bits)
0x1A iDAC 2 value (low 5 bits)
0x1B voltage bias generator control (low 7 bits)
0x1C bandgap control (low 6 bits)
0x1D bandgap bias tuning (low 5 bits)
0x1E voltage bias tuning (sink) (low 6 bits)
0x1F voltage bias tuning (source) (low 5 bits)
0x20 to 0x37 Input digital data bus assignments (low 4 bits):
0x38 User digital read-back (sampled)
0x40 to 0x4B Output digital data bus assignments (low 4 bits):
0x50 Selected project (low 5 bits)
0x51 User project configuration
0x52 User project bias configuration

Note: digital pins can be either input or output. If both are enabled, then pin is configured as an output.

Project selection includes 16 projects + one diagnostic + one setting to disable all projects. The diagnostic mode connects the biases to the shared analog pins and loops digital inputs back to digital outputs.

Each SPI command is followed by a 1-byte address, indicating an SPI register or SRAM address. In the case of SRAM addresses, only addresses 0 to 255 are individually addressible. Addresses up to 1023 are accessible by continuous reading or writing with auto-increment.

Note: SRAM cannot be accessed while the sequencer is running.

In the absence of a clock, the sequencer cannot run, but the digital input bus values can still be set to constants or connected to digital input pins.

Sequencer mode values

Value 0x0 = Binary up count
Value 0x1 = Binary down count
Value 0x2 = Grayscale up count
Value 0x3 = Grayscale down count
Value 0x4 = Pseudorandom LFSR output
Value 0x5 = Set all sequencer bits to zero
Value 0x6 = Set all sequencer bits to one

SRAM mode values

Value 0x0 = Increment at sequencer clock rate
Value 0x1 = Increment at variable clock rate
Value 0x2 = Increment at sequencer end

Clock prescaler values

Value 0 to 255 (0 = divide by 256)

Input digital bus bit assignments

Value 0x0 to 0xb = digital input pin 0 to 11
Value 0xd = constant 0
Value 0xe = constant 1
Value 0xf = sequencer output (address 0x20 to 0x2f) or SRAM output (address 0x30 to 0x37)

Output digital bus bit assignments

Value 0x0 to 0xb = digital output pin 0 to 11

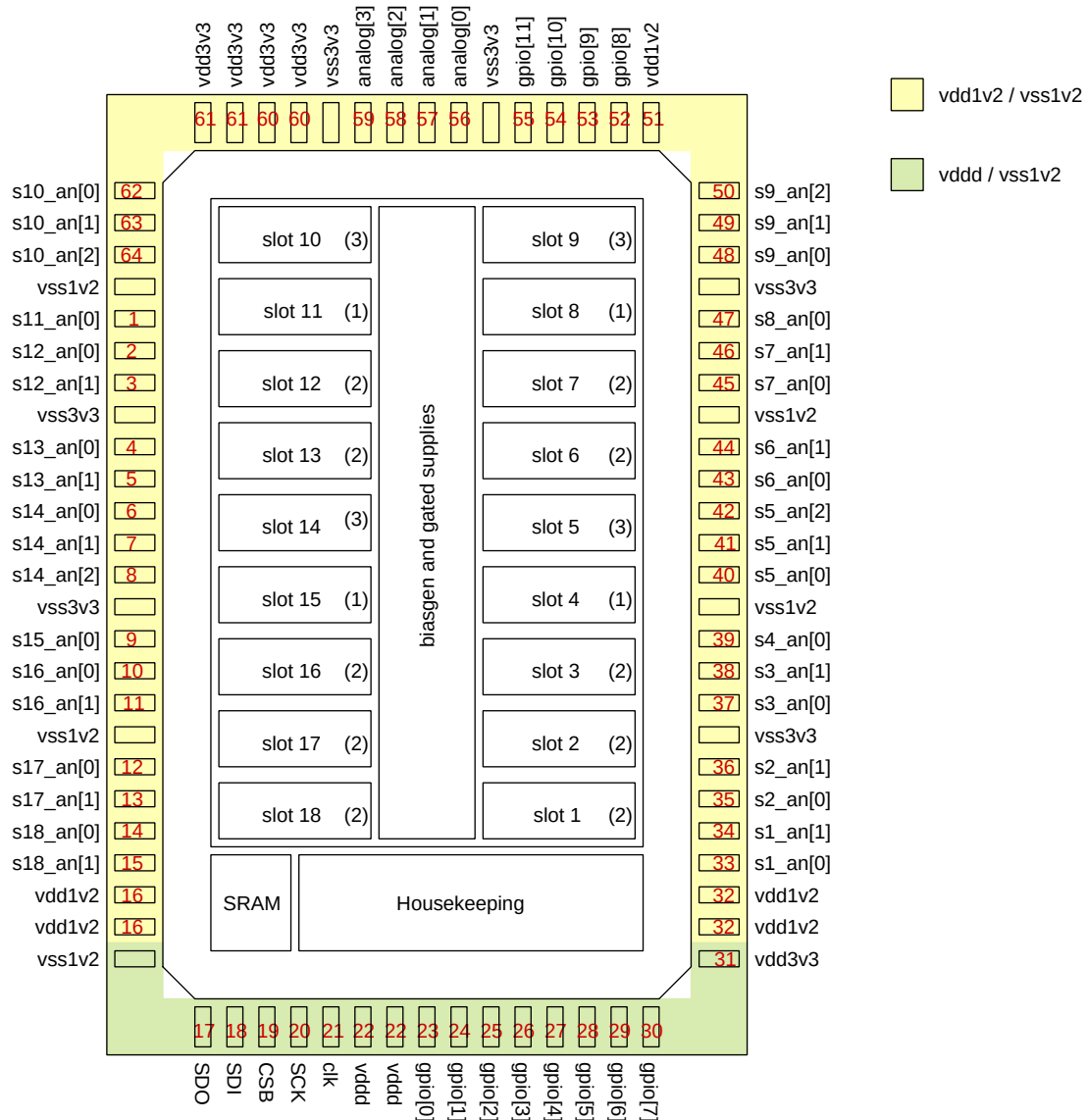
User project configuration bits

Bit 0 = project enable
Bit 1 = 3.3V domain power switch
Bit 2 = 1.2V domain power switch
Bit 3 = digital bus enable
Bits 4-7 = analog bus enables

User project bias configuration bits

Bit 0 = ibias1 switch
Bit 1 = ibias2 switch
Bit 2 = vbiase switch

Analog harness 1.2V power domains

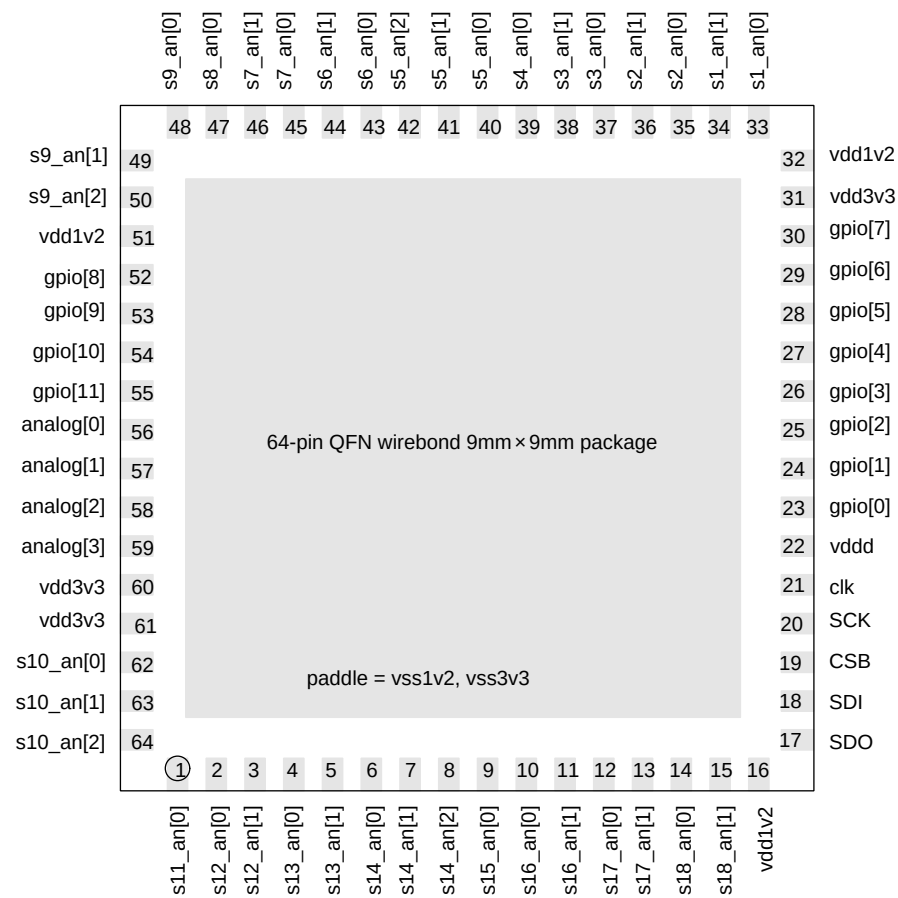


NOTE: There is only one 3.3V power domain which is continuous around the padframe.

There is no way to make the housekeeping 1.2V domain ground return separate from the user 1.2V domain ground return.

Pins marked in red are QFN package pins (see bonding diagram). All ground pins are downbonded to the QFN paddle. Each power supply has one double-bonded pin. Each project with one or more dedicated pins has a pin adjacent to a ground return. Projects requiring proximity to a specific ground return must be assigned to an appropriate slot.

Analog harness 1.2V QFN package pinout

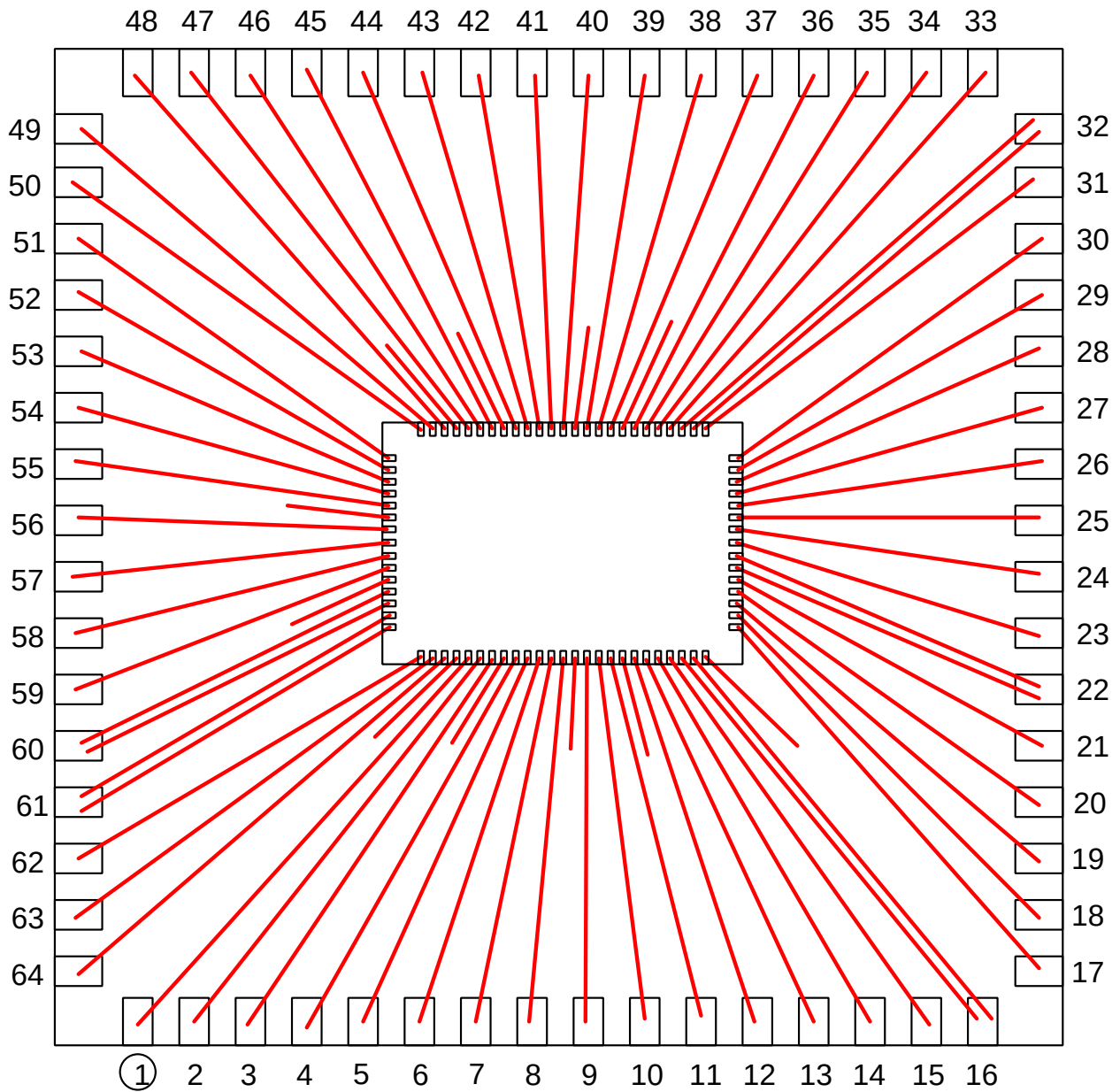


Process: IHP sg13cmos5l project

Project: sg13cmos5l_ocd_openframe (80 pin padframe)

64-pin QFN wirebond 9mm×9mm package

Die size: 2.10 mm×3.14 mm (rotated -90 degrees)



Bond drawing: version 2.1 August 3, 2026 Open Circuit Design LLC